

Abhishek Mane

abhishek.mane.work@gmail.com | +91-7020870063 | Ahmedabad, Gujarat, India
Portfolio: abhishekmane6122.github.io/my-portfolio | github.com/abhishekmane6122
linkedin.com/in/abhishek-mane-aiml | substack.com/@abhimane

EDUCATION

- Terna Engineering College** Osmanabad, India
Bachelor of Technology – Artificial Intelligence and Data Science Dec. 2021 – 2025

EXPERIENCE

- CognitBotz Solutions** Ahmedabad, India
Artificial Intelligence Engineer – Client: Adani Group Sep. 2025 – Present
 - Aegis – Enterprise Compliance Platform:** Designed and developed a full-stack, production-grade compliance intelligence platform for Adani Group's Secretarial & Legal teams. Built six independently deployable modules – BSE Intelligence, RBI Compliance, SEBI Hub, Director Disclosure Management, Insider Trading Surveillance, and Minutes Preparation – behind a unified React UI. Platform saved **250+ FTE hours per quarter** and is live in production.
 - Regulatory RAG Pipeline (BSE, SEBI, RBI):** Engineered automated scraping and ingestion pipelines (using Playwright and Requests) that collect daily notifications from BSE, SEBI, and RBI portals, extract content from PDFs via PyMuPDF and OCR, and generate AI summaries using Phi-3-mini LLM. Indexed documents into ChromaDB vector store for semantic search and pushed structured data to Azure PostgreSQL. Delivered automated daily email reports via SMTP. Eliminated **6+ hours of manual daily monitoring** and reduced SEBI notification noise by **71%**.
 - Multi-Agent NL-to-SQL System:** Designed a multi-agent architecture where a routing agent classifies user intent and delegates to either a deterministic query handler or a GPT-powered dynamic SQL generation agent. Engineered prompt templates with schema context injection for SAP HANA-compatible SQL generation. Built a query validation layer to sanitize LLM output, prevent SQL injection, and enforce read-only access.
 - Equity Pulse – Market Analytics Platform:** Built an end-to-end equity analytics platform for Adani Group's finance and portfolio teams. Designed ETL pipelines on a scheduled VM to extract, validate, and load daily bhavcopy data from BSE and NSE into Azure PostgreSQL. Developed a backend computation engine to calculate moving averages, volatility metrics, delivery ratios, and relative performance indices. Built the React frontend with interactive dashboards, candlestick charts, competitor analysis, and shareholding pattern tracking across **7 Adani business units** and **3.9M+ investor records**. Eliminated **4–6 hours of daily manual analyst work**.
 - Platform Infrastructure, AI Observability & RAG Evaluation:** Built a custom **AI Observability and Evaluation Platform** monitoring all production agents and RAG pipelines in real time. Integrated **RAGAS** and **DeepEval** frameworks to benchmark retrieval context precision, answer faithfulness, semantic recall, and LLM hallucination rates across document ingestion workflows. Deployed centralized telemetry dashboards displaying distributed execution traces, component downtime, token consumption, p95/p99 latency, and error spikes. Engineered an **autonomous L1 AI Support Engineer agent** on top of system log streams that parses error traces, performs real-time root-cause analysis (RCA), and automatically generates diagnostic quick-fix playbooks, reducing MTTR and manual triage effort. Enforced enterprise security via Azure AD SSO with RBAC, Key Vault, and TLS/AES-256 encryption.
 - Stakeholder Management & Delivery:** Directly engaged in stakeholder management with Adani Group leadership – conducting project progress updates, requirements gathering sessions, and sprint reviews to ensure delivery alignment with business objectives.
- CognitBotz Solutions** Hyderabad, India
Artificial Intelligence Engineer Nov. 2024 – Sep. 2025
 - LLM Engineering & Deployment:** Developed and fine-tuned LLMs using LangChain and Agno with few-shot and transfer learning. Designed APIs for LLM integration and deployed production chatbots, Q&A systems, and web agents handling text, documents, and multi-modal inputs.
 - AI Evaluation & Responsible AI:** Evaluated LLM performance on accuracy, relevance, and fairness metrics. Implemented bias mitigation, data privacy safeguards, and responsible AI practices to prevent misuse and harmful content generation.
- Infosys Springboard** India
Artificial Intelligence Developer (Internship) Sep. 2024 – Dec. 2024
 - AI Application Development:** Built AI/ML applications under the Infosys Springboard program, gaining hands-on experience with enterprise AI workflows and NLP pipelines.
- Cyber Crime Police Station – India** Osmanabad, India
Data Analyst Oct. 2023 – Jan. 2024
 - Data Management:** Managed user credentials for the National Cyber Crime Reporting Portal. Extracted and organized complaint data in Excel to improve case handling efficiency. Maintained portal by updating complaint statuses and coordinating directly with victims on case resolutions.

PROJECTS

- **Aegis – Enterprise Compliance Intelligence Suite** | *CognitBotz, Client: Adani Group*: Production-grade multi-module platform automating BSE/SEBI/RBI regulatory monitoring, director disclosure management, insider trading surveillance across 3.9M+ investor records, and board meeting minutes generation. Built automated scraping with Playwright/PyMuPDF, ChromaDB semantic vector search, and daily automated SMTP reports, saving **250+ FTE hours/quarter**. *Stack: React, FastAPI, Python, LangChain, LangGraph, Azure OpenAI, Cohere Embeddings, pgvector (Azure PostgreSQL), Azure Key Vault, Azure VM, Nginx, Docker, GitLab.*
- **Ward – Zero-Trust AI Model Security & Static Opcode Ingestion Gateway** | *Enterprise SecOps & MLOps Architecture*: Zero-trust security gateway for open-source AI models in regulated enterprises. Engineered C-level static bytecode disassembler (`pickletools.genops`) intercepting **100% of PyTorch/Pickle malware** (<350ms latency) without execution, Protocol 4 `STACK_GLOBAL` lookback buffers, magic-byte sniffing (`.pt`, `Safetensors`, `GGUF`), Jaccard word-bigram shingle deduplication (0.40 threshold), Hardware Resource Sizing Advisor (FP32/FP16/INT8/INT4 VRAM across A100/H100/L40), and CycloneDX AIBOM generation for air-gapped GPU deployments aligned with EU AI Act and NIST AI RMF. *Stack: Python 3.13, FastAPI, React 18, TypeScript, SQLAlchemy 2.0, SQLite/PostgreSQL, Groq/OpenRouter (LLaMA-3.3-70B), Pydantic v2, Tailwind CSS.*
- **Naipunya – Low-Latency Voice AI & Multi-Agent Actuarial Advisory System** | *InsurTech & Speech Systems Architecture*: Voice-first conversational multi-agent system for Indian insurance advisory. Engineered client-side Web Audio API RMS silence detection (0ms network VAD delay), speculative parallel execution (`asyncio.gather`) running headless Selenium web RAG and LLM reasoning concurrently, progressive 7-slot state machine with sliding-window memory, dual deployment (Cloud Groq LPU vs **100% offline edge CPU inference** via Ollama IBM Granite 2B, Faster-Whisper INT8, Kokoro-82M ONNX in <4 GB RAM), and deterministic actuarial tools (<1% hallucination rate) across English, Hindi, and Telugu. *Stack: FastAPI, React 19, Python, Groq LPU, Llama-3.3, Qwen-2.5, Ollama (IBM Granite), Faster-Whisper, Kokoro ONNX, Edge-TTS, Web Audio API, Selenium, Langfuse.*
- **PerkAI – Autonomous Multi-Agent Capital Discovery & Combinatorial Stacking Engine** | *FinTech & Agentic Systems Architecture*: Autonomous multi-agent platform discovering **\$180k–\$340k in startup credits** and non-dilutive government subsidies in <18s. Engineered domain crawler with headless Chrome SPA rendering, algorithmic Grounding Guard enforcing character-window quote verification (0.0% ungrounded claims), in-process deterministic JSON rule gating (78% token reduction), constrained combinatorial stacking optimizer for mutually exclusive cloud credits, in-process 2048-dim vector math (NumPy cosine similarity on PostgreSQL float[]), and distributed AI gateway with automated failover. *Stack: Python, FastAPI, React 19, PostgreSQL (JSONB), NVIDIA NIM (Nemotron-3 2048-dim), Groq, OpenRouter, Langfuse v3, Prometheus, Grafana, Loki, SSE.*
- **Equity Pulse – Enterprise Equity Analytics Platform** | *CognitBotz, Client: Adani Group*: End-to-end equity analytics platform tracking 5 Adani business units with automated ETL from BSE/NSE, real-time dashboards, candlestick charts, competitor analysis, and shareholding pattern portal across **3.9M+ investor records**. Eliminated 4–6 hrs of daily manual analyst work. *Stack: React, TypeScript, FastAPI, Python, Azure PostgreSQL, pgvector, Azure VM, APScheduler, Recharts, Docker, GitLab.*
- **Quantus Med – Multimodal AI Clinical Diagnostic Intelligence** | *CognitBotz*: Clinical AI platform fusing medical imagery (DICOM/JPG/PNG) with real-time physician voice dictation to generate structured SOAP diagnostic reports in under **500ms**. Uses Llama-3 and Gemini 1.5-Flash for vision, Whisper-large-v3 for transcription, and Edge-TTS for hands-free playback. HIPAA/GDPR-compliant zero-knowledge inference. *Stack: React-Vite, FastAPI, LangChain, Groq LPU, Llama-3, Gemini 1.5-Flash, Whisper-large-v3, Edge-TTS.*
- **Doc Capture – Intelligent Document Extraction API** | *CognitBotz*: FastAPI backend for automated extraction and export of structured data from multi-type documents (Aadhaar, PAN, Driving License, Voter ID, invoices). Supports template-based invoice mapping, batch processing, and Excel/JSON export. Docker Compose deployed. *Stack: Python, FastAPI, Groq LLM, PyMuPDF, Camelot, Openpyxl, Docker.*
- **OmniQA-Agent – Autonomous Requirements & Test Engineering Orchestrator** | *Personal*: Autonomous QA platform crawling websites via Playwright headless Chromium, parsing DOM with BeautifulSoup, and orchestrating LangChain pipelines through Groq Llama-4 to auto-generate user stories, test plans, and test cases. Exports color-formatted Excel workbooks ready for Jira/ALM integration. *Stack: Python, Streamlit, Playwright, BeautifulSoup4, LangChain, Groq (Llama-4), Openpyxl.*

CORE STACK

- **AI / LLMs**: Large Language Models (OpenAI, Azure OpenAI, Llama, Claude, Gemini), Fine-Tuning & Integration, LangChain, LangGraph, AI Orchestration, Multi-Agent Architectures, Prompt Engineering
- **RAG & Search**: Retrieval-Augmented Generation (RAG), Semantic Search, Azure AI Search, Weaviate, FAISS, pgvector (Azure PostgreSQL), ChromaDB
- **Backend & APIs**: Python, FastAPI, Flask, REST APIs & Microservices for AI Applications, PostgreSQL, SQLite, Async Programming
- **MLOps & Cloud**: MLOps (Model Deployment, Monitoring, Evaluation & Quality Tuning), Containerization (Docker, Kubernetes), Azure Cloud Platform, CI/CD, Git, GitLab

AWARDS & RECOGNITION

- **Performance Award** (December 2025), Adani Group: Recognized by leadership and stakeholders for designing and delivering impactful AI-driven portals and solutions. Work was highly appreciated and received positive feedback across multiple teams at the organization.

CERTIFICATIONS

- CCSK v4.1 Foundation Training – Cloud Security | Cloud Computing Fundamentals | AICTE-Cisco Cyber Security Virtual Internship | Google for Developers Cloud Bootcamp (GeeksforGeeks)